

Brianna Cheng

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(917)-968-2350

EDUCATION

Cornell University, College of Engineering, Ithaca, NY

Expected Graduation May 2027

Major: Mechanical Engineering

Relevant Courses: Fluid Mechanics, System Dynamics, Mechanics of Materials, Dynamics, Introduction to Mechanical Design, Statics & Mechanics of Solids, Thermodynamics, Introduction to Computing: Design and Development

PROFESSIONAL EXPERIENCE

Course TA—Introduction to Mechanical Engineering

August 2025 – Present

- Led office hours and review sessions for 30+ students to prepare for exams and improve comprehension of force and pressure analysis, thermodynamics, and gear systems
- Facilitated in-lecture problem solving by reinforcing students' understanding of core mechanical fundamentals

Teradar, Summer Mechanical Internship at Sensor Company

June 2025 – August 2025

- Designed and machined custom mounts and enclosures to house sensors for dynamic testing across automobile and aerial platforms (cars, go-carts, helicopter, robot arms, and gimbals) and in lab, enabling faster data collection expanding the range of testing environment
- Assembled and optimized sensor systems by reworking thermal and mechanical components—heat spreaders, heatsinks, counter sink holes—to enhance cooling efficiency and device performance
- Collaborated with cross disciplinary engineers to refine mount designs and streamline testing

ENGINEERING EXPERIENCE

Autonomous Boat Project Team, Cornell University

February 2024- Present

- Engineered and constructed custom cooling system, battery enclosure, and buoy guards for the team's autonomous boat competing in the annual RoboBoat International Competition
- Fabricated boat hulls, almas, and bridge decks using composite materials and custom CNC- milled molds coordinating with teammates to ensure structural accuracy and hydrodynamic efficiency
- Applied CAD and FEA (Ansys, Fusion 360) to evaluate components' performance and guide design iterations

Cornell Autonomous Motion, Cornell University

January 2025 - Present

- Modeled and analyzed structural components to improve balance and stability of an autonomous bike prototype
- Developed digital prototypes and structural frameworks for a quadrupedal robot dog

Cornell Engineering Entertainment & Design

January 2025 - Present

- Collaborated on multidisciplinary interactive installations integrating mechanics and electronics
- Designed sliding gear mechanism for animatronic axolotl and rotating obstacle course for Flappy Bird machine

Extracurriculars

Alpha Pi Omega (Community Service Fraternity)

August 2024 - Present

Cornell Pickleball Club

August 2025 – Present

Cornell Media Guild:

August 2025 - Present

- Maintain and troubleshoot radio broadcasting to ensure uninterrupted radio transmission and hardware reliability

SPECIALIZED SKILLS

Programs: Solidworks, Fusion 360, Ansys, Orca, MATLAB Onshape, Squarespace, Click-Up, and Microsoft Office

Fabrication Skills: CAD printing, CNC machining (drill press, mill, and handheld tools), composite lay-up (fiberglass, epoxy) metal working, woodworking and polymer processing

Core Skills: AutoCAD modeling, Prototyping & Testing, System & Design integration, Design for Manufacturability, Thermal Management, Mechanical Assembly